

Architecture

Ansible provides sysadmins with a strong IT automation engine that automates cloud provisioning, configuration management, application deployment, intra-service orchestration and other IT needs.

- **Modules:** Ansible connects to all nodes available in a network using Ansible Modules. These are programs regarded as resource models of the system state. The Modules library can reside on any machine and there is no requirement for any server, daemon and database.
- **Plugins:** These small pieces of code build up the entire functionality.
- **Inventory:** This consists of all machines managed by Ansible.

- **Playbooks:** These orchestrate multiple slices of the infrastructure topology.

Figure 3 illustrates the architecture of Ansible.

Features

The features of Ansible include the following:

- It manages the system with a collection of scripts and ad hoc practices in a very simple manner.
- It offers simple solutions for all configuration management issues. It is designed to be minimal in nature, secure, consistent and easy-to-learn for administrators and developers.
- Ansible has a state-driven resource model that describes the desired state of computer systems and

services, no matter what state a system is in. Ansible transforms it to the desired state, which allows reliable and repeatable IT infrastructure configuration, avoiding potential failures from scripting and script-based solutions.

- Ansible relies on the most secure remote configuration management system available, as its default transport layer is OpenSSH. It delivers all modules to remote systems and executes tasks, as needed, to enact the desired configuration. These modules run with user-supplied credentials, including support for sudo and even Kerberos, and clean up on their own when the job is complete.
- It consists of 1300+ modules and an active community for support and development.

Official website: <https://www.ansible.com/>

Latest version: Ansible Tower

Juju Charms

Juju is an open source model for service-oriented architecture and deployment. It is a powerful service orchestration tool from Ubuntu that helps systems administrators to define, configure and deploy services to any cloud or bare metal system. A Juju Charm is a structured bundle of YAML configuration files and scripts for a software component that enables Juju to deploy and manage the software component as a service, according to best practices. It is mainly programmed in Go, with 99 per cent of the code composed in this language.

Charms are composed of metadata, configuration data, and hooks with some extra support files. Charms use hooks, typically written as shell scripts, which Juju invokes to manage the life cycle of the service. There are two types of hooks—Unit hooks and relation hooks. The former include *install*, *config-changed*, *start*, *upgrade-charm* and *stop*. Relation hooks include *joined*, *changed*, *departed* and *broken*. Defining relationships in a Charm simplifies deployment by embedding the logic of what the Charm can and cannot connect to.

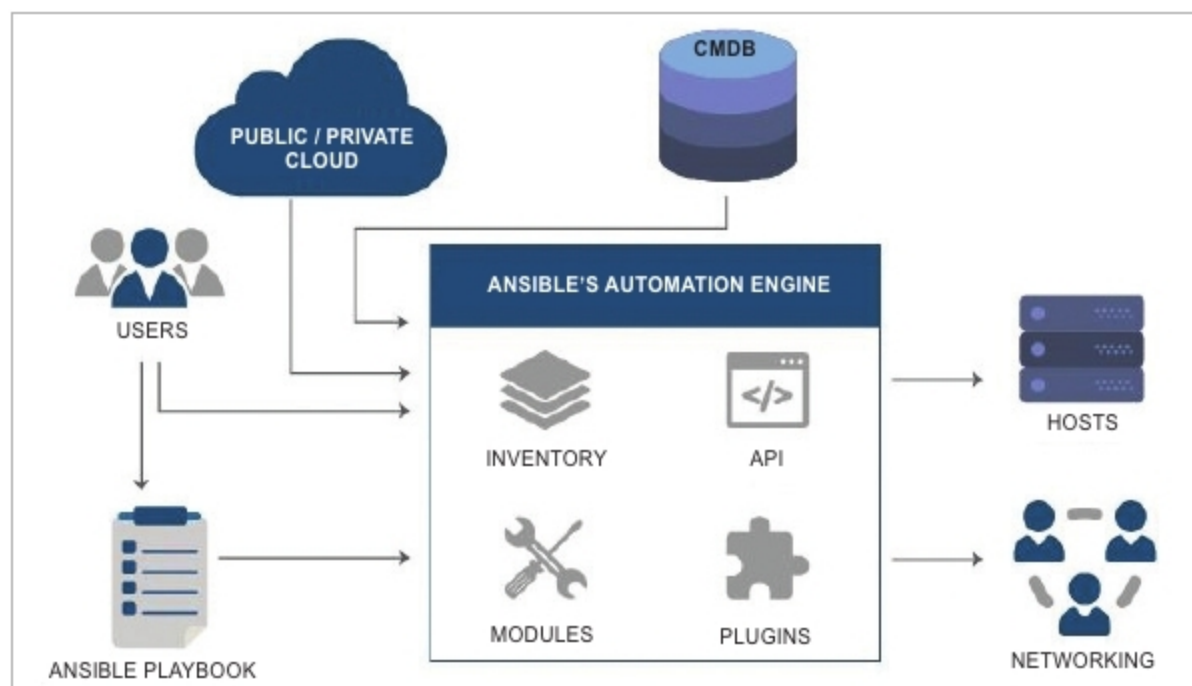


Figure 3: Ansible architecture

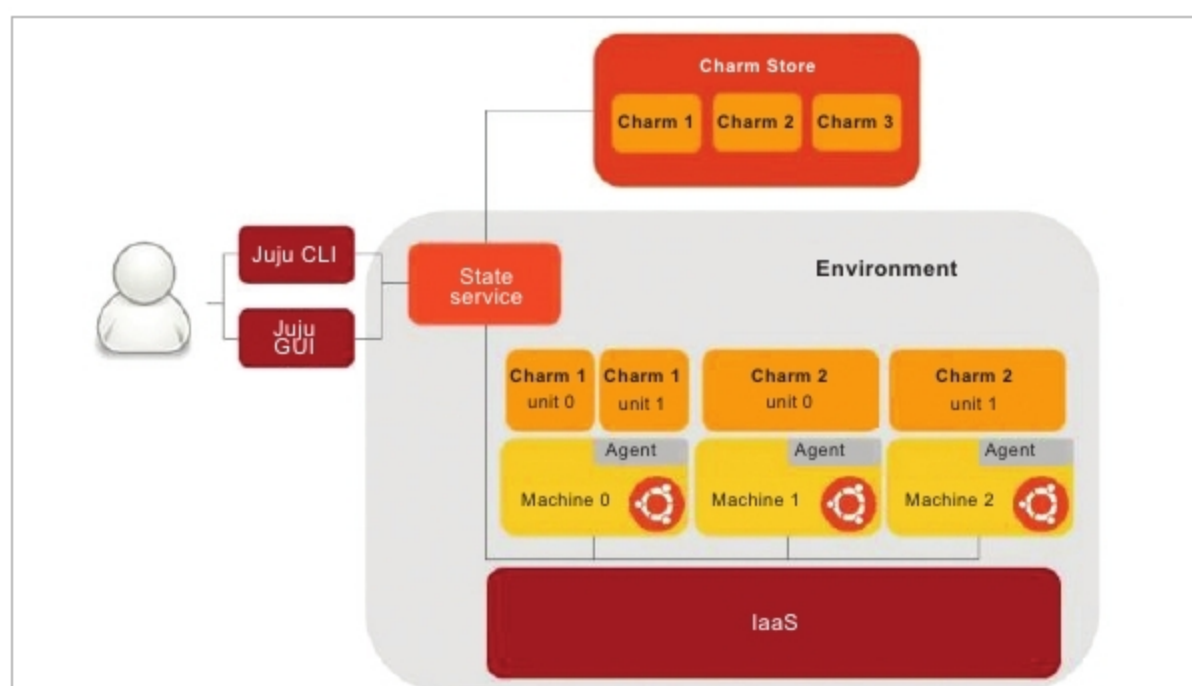


Figure 4: Juju Charms architecture